

Graph Neural Network Tutorial – 1

- **What is the primary purpose of a Graph Neural Network (GNN)?**
 - A) Image classification
 - B) Text generation
 - **C) Learning from graph-structured data** //
 - D) Speech recognition
- **In a GNN, a node's feature is updated based on:**
 - A) Only its initial feature
 - **B) Its neighboring nodes' features**
 - C) Random noise
 - D) A pre-defined constant
- **Which of the following is a key advantage of GNNs?**
 - A) Their ability to work on grid-like data
 - **B) Their invariance to node ordering**
 - C) They are only used for social network analysis
 - D) They do not require any training data
- **Which of the following is NOT a typical application of GNNs?**
 - A) Molecular property prediction
 - B) Social network analysis
 - **C) Speech synthesis**
 - D) Recommender systems
- **Which GNN operation allows a node to aggregate information from its neighbors?**
 - A) Forward propagation
 - B) Backward propagation
 - **C) Message passing**
 - D) Loss computation
- **In Graph Convolutional Networks (GCNs), convolution occurs over:**
 - A) Pixels in an image
 - B) Time steps in a sequence
 - **C) Nodes and their neighbors in a graph**

- D) Hidden states in an RNN
- **What is a typical way to represent a graph in a GNN?**
 - A) Adjacency matrix
 - B) One-hot encoding
 - C) Fourier transform
 - D) Activation map
- **Which type of pooling is often used in GNNs to reduce the graph size?**
 - A) Max pooling
 - B) Sum pooling
 - C) Average pooling
 - D) Graph coarsening
- **The process of aggregating features in a GNN is referred to as:**
 - A) Convolution
 - B) Propagation
 - C) Pooling
 - D) Embedding
- **What is the main challenge addressed by GNNs compared to other neural networks?**
 - A) Handling structured data like images
 - B) Learning from non-Euclidean data such as graphs
 - C) Training deep architectures efficiently
 - D) Enhancing transfer learning across domains